

Time: Three Hours

Maximum Marks: 75

SECTION A – (10 × 2 = 20 marks)**Answer the following questions**

1. Write down the expansion for $(1 - x)^n$.
2. Write Maclarin's series for e^x .
3. Define symmetric matrix.
4. Find the values of a,b,c if $\begin{pmatrix} a+3 & 3a-2b \\ -3a-c & a+b+c \end{pmatrix} = \begin{pmatrix} 2 & -7+2b \\ b+4 & 2a \end{pmatrix}$
5. State Cayley – Hamilton theorem.
6. Prove that $\Delta = E - 1$.
7. Prove that $1 - \nabla = E^{-1}$.
8. Write down the expansion of $\cos \theta = \log(x + \sqrt{x^2 - 1})$.
9. Prove that $e^{i\theta} = \cos \theta + i \sin \theta, \forall \theta$.
10. Prove that $\cosh^{-1} x = \log(x + \sqrt{x^2 - 1})$.
11. If $x = r \cos \theta; y = r \sin \theta$, Find the value of $\frac{\partial(x,y)}{\partial(r,\theta)}$.
12. Find the n^{th} derivative of $\left(\frac{1}{(ax+b)}\right)$.

SECTION B – (5 × 5 = 25 marks)

Answer any FIVE questions

13. Find the sum to infinity of the series $1 - \frac{1}{8} + \frac{1.3}{8.16} - \frac{1.3.5}{8.16.24} + \dots \infty$.

14. Prove that $a^x = 1 + x \log_e a + \frac{x^2}{2!} (\log_e a)^2 + \dots$

15. Show that the matrix

$$A = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{pmatrix} \text{ is orthogonal.}$$

16. Find the characteristics equation of the matrix

$$\begin{pmatrix} 0 & a & b \\ A & 0 & c \\ B & c & 0 \end{pmatrix}$$

17. Expand $\tan 6\theta$ in terms of $\tan \theta$.

18. If $x = u(1 + v)$ and $y = v(1 + u)$, find $\frac{\partial(x,y)}{\partial(u,v)}$.

19. Show that the minimum value of $f = xy + a^3 \left(\frac{1}{x} + \frac{1}{y} \right)$ is $3a^2$.

SECTION C – (3 × 10 = 30 marks)

Answer any THREE questions

20. Show that $\log \frac{n+1}{n-1} = \frac{2n}{n^2+1} + \frac{1}{3} \left(\frac{2n}{n^2+1} \right)^3 + \frac{1}{5} \left(\frac{2n}{n^2+1} \right)^5 + \dots$

21. Verify Cayley Hamilton theorem and hence find the inverse

of the matrix. $\begin{pmatrix} 1 & 2 & -1 \\ 3 & -3 & 1 \\ 2 & 1 & -2 \end{pmatrix}$

22. Using Newton's forward interpolation formula find y at $x=8$ from the table.

x	0	5	10	15	20	25
y	7	11	14	18	24	32

23. Prove that $\cos^5 \theta = \frac{1}{6} [\cos 5\theta + 5\cos 3\theta + 10\cos \theta]$.

24. If $y = (x + \sqrt{1+x^2})^m$ prove that $(1+x^2)y_{n+2} + (2n+1)xy_n + (n^2 - m^2)y_n = 0$.